

What we're doing

A no-folding null model for contact prediction: for each of the 554 eval proteins, mmseqs-align it against all 4.13M contacts-v1 training documents, copy the top-k neighbors' ground-truth contacts through the alignment, vote, and score R-precision with the exact same metric harness as every other predictor.

Why

To test how much of MarFold's contact-prediction skill is just memorizing training folds. If copying nearest neighbors matches the model, it's largely retrieval.

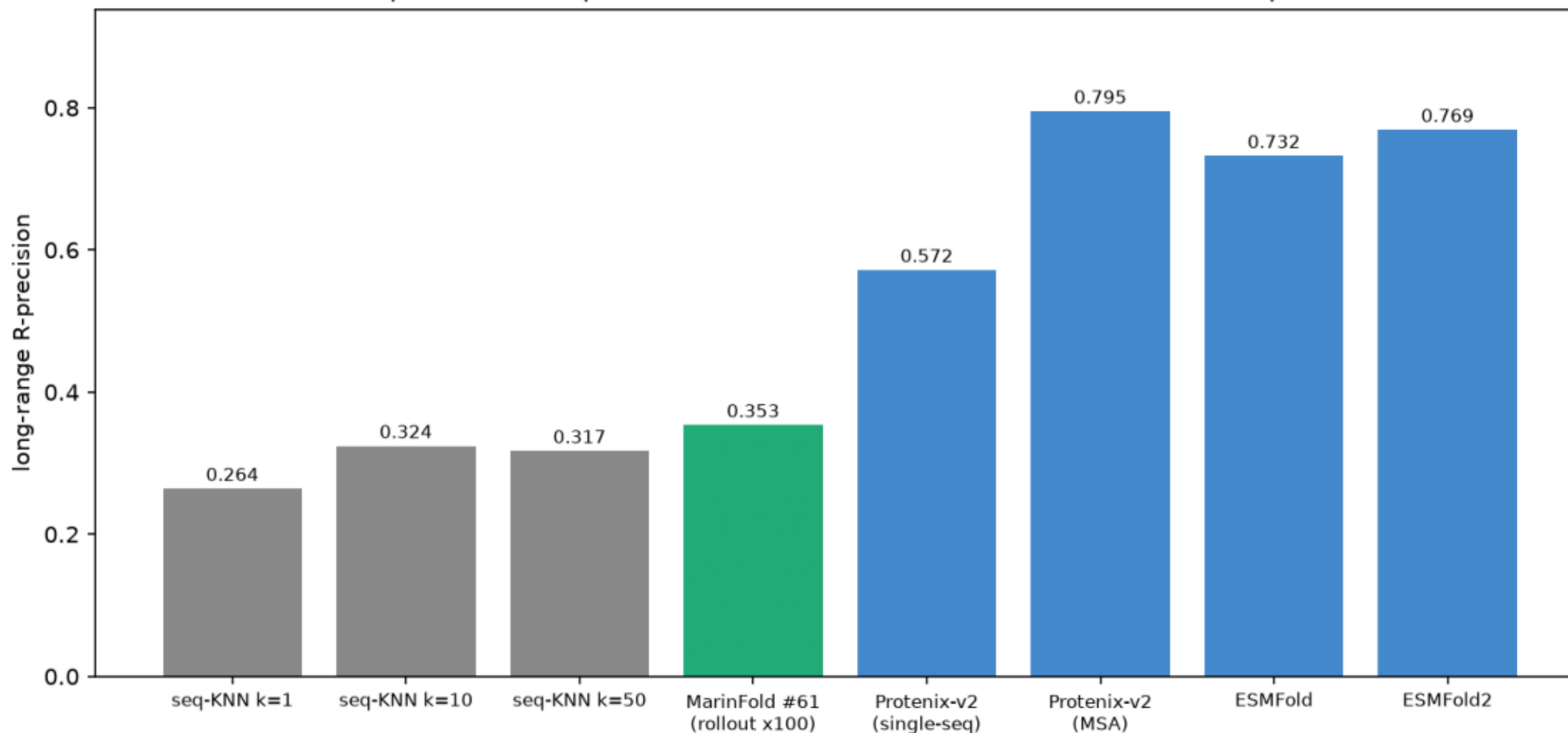
Results so far

seq-KNN $k=10$ ties MarFold #61 on AVERAGE long-range R-precision (0.324 vs 0.353) — but that's a coincidence of averaging. Stratified by nearest-neighbor identity, seq-KNN rises monotonically (0.00 $\rightarrow$ 0.69) while MarFold stays flat ($\sim$ 0.35) and is the only predictor that works on the 139 no-homolog proteins. So MarFold is NOT just memorizing: its accuracy is independent of training-sequence proximity.

headline_rprecision.png

Mean long-range R-precision over the 554 eval proteins, all scored under exp89's metric harness. The issue's success criterion: seq-KNN k=10 (0.324) ties MarInFold #61 (0.353) — a no-folding copy-the-neighbours baseline nearly matches the LM on average.

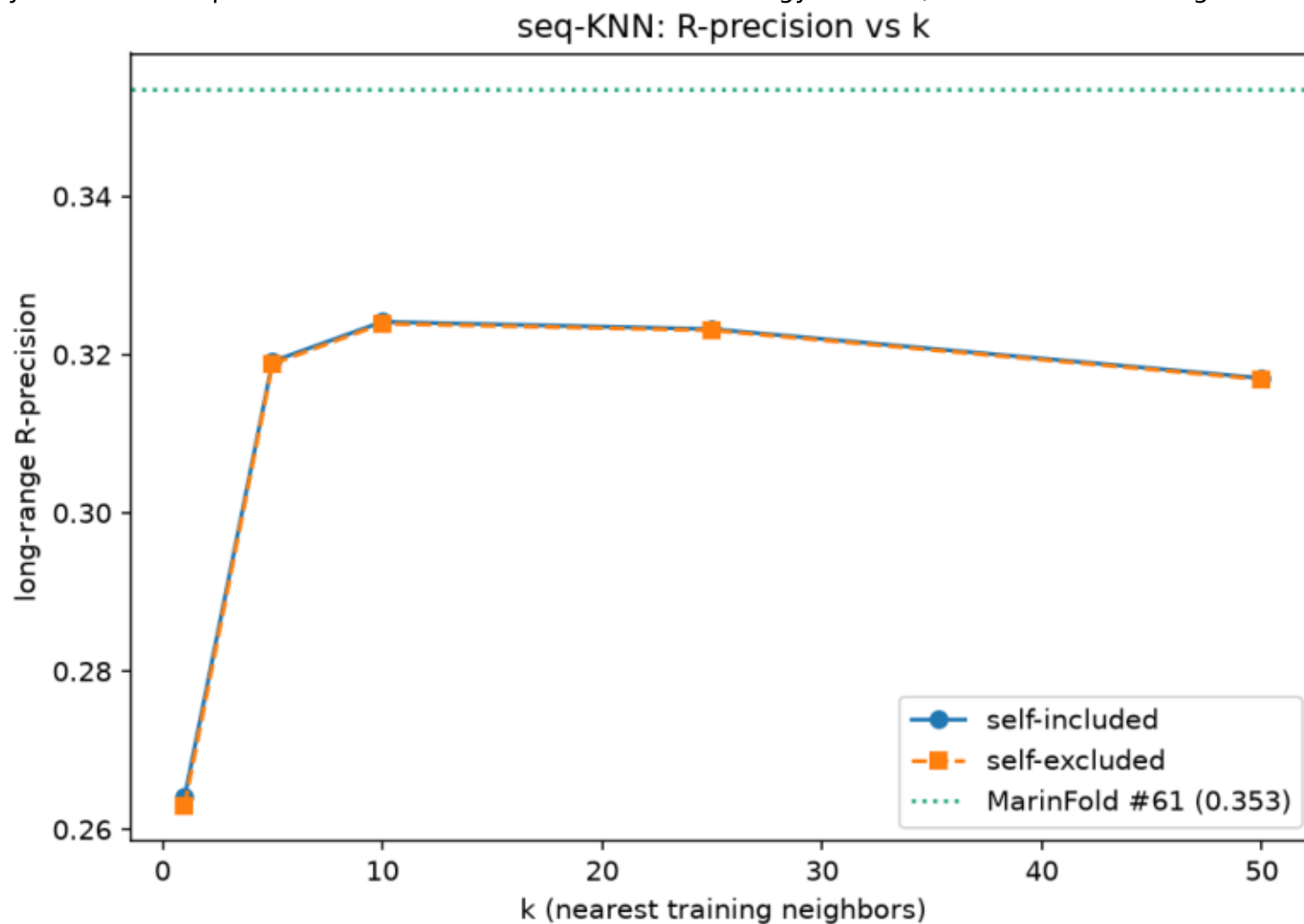
Contact prediction: sequence-KNN null model vs MarInFold and structure predictors



```
$ plot_knn.py --base-csv ../exp82_evals_contacts_v1_contact_prediction/_scratch/contact_precision_with_rollout.csv
```

k_sweep.png

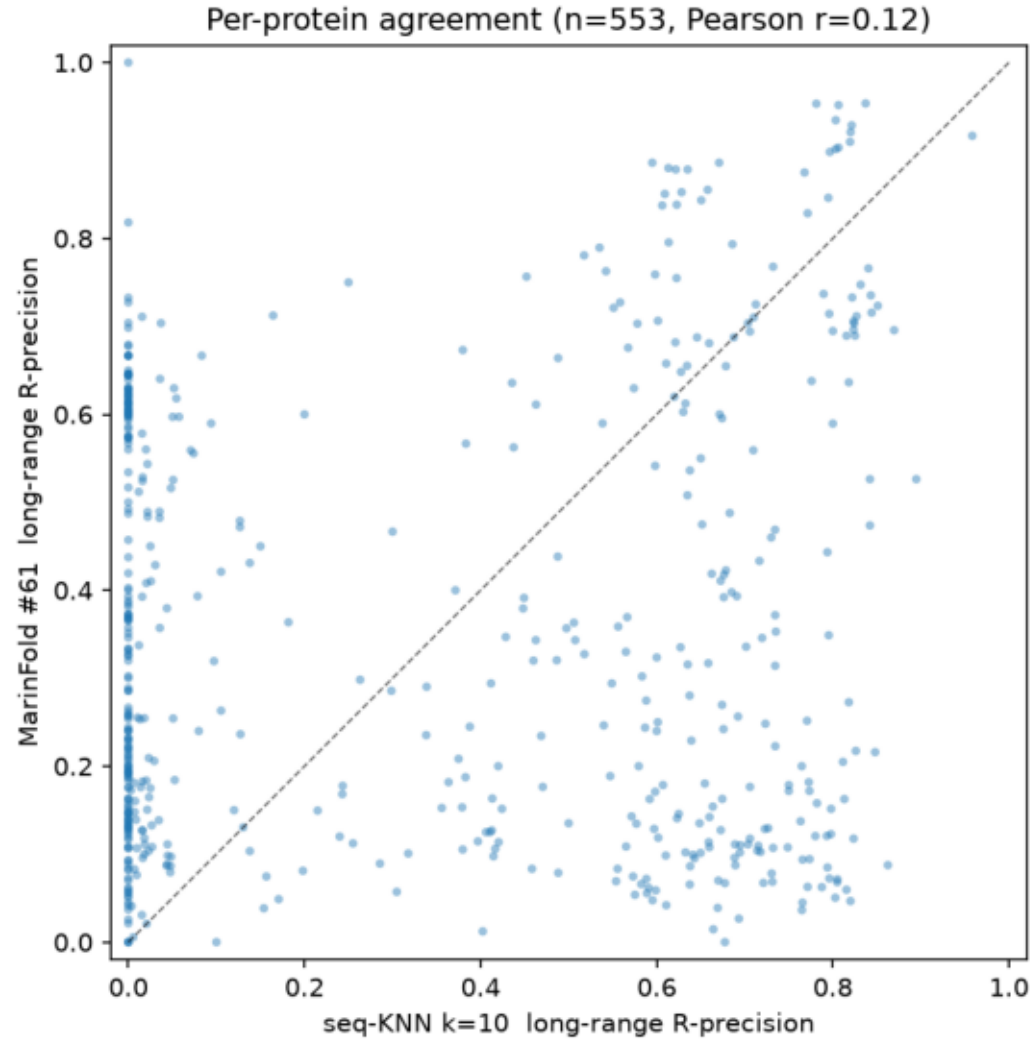
seq-KNN long-range R-precision vs k (nearest training neighbours voted), self-included vs self-excluded. Peaks at k=10; the two curves coincide because only 2 of 554 eval proteins have a verbatim self-hit — homology transfer, not train/eval leakage.



```
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```

memorization_scatter.png

Per-protein long-range R-precision: seq-KNN k=10 (x) vs MarFold #61 (y); dashed line = parity. The cloud does not hug the diagonal (Pearson $r=0.12$) — the two are accurate on different proteins, so their matching averages are coincidental.



```
$ plot_knn.py --base-csv ../exp82_evals_contacts_v1_contact_prediction/_scratch/contact_precision_with_rollout.csv
```

rprecision_vs_identity.png

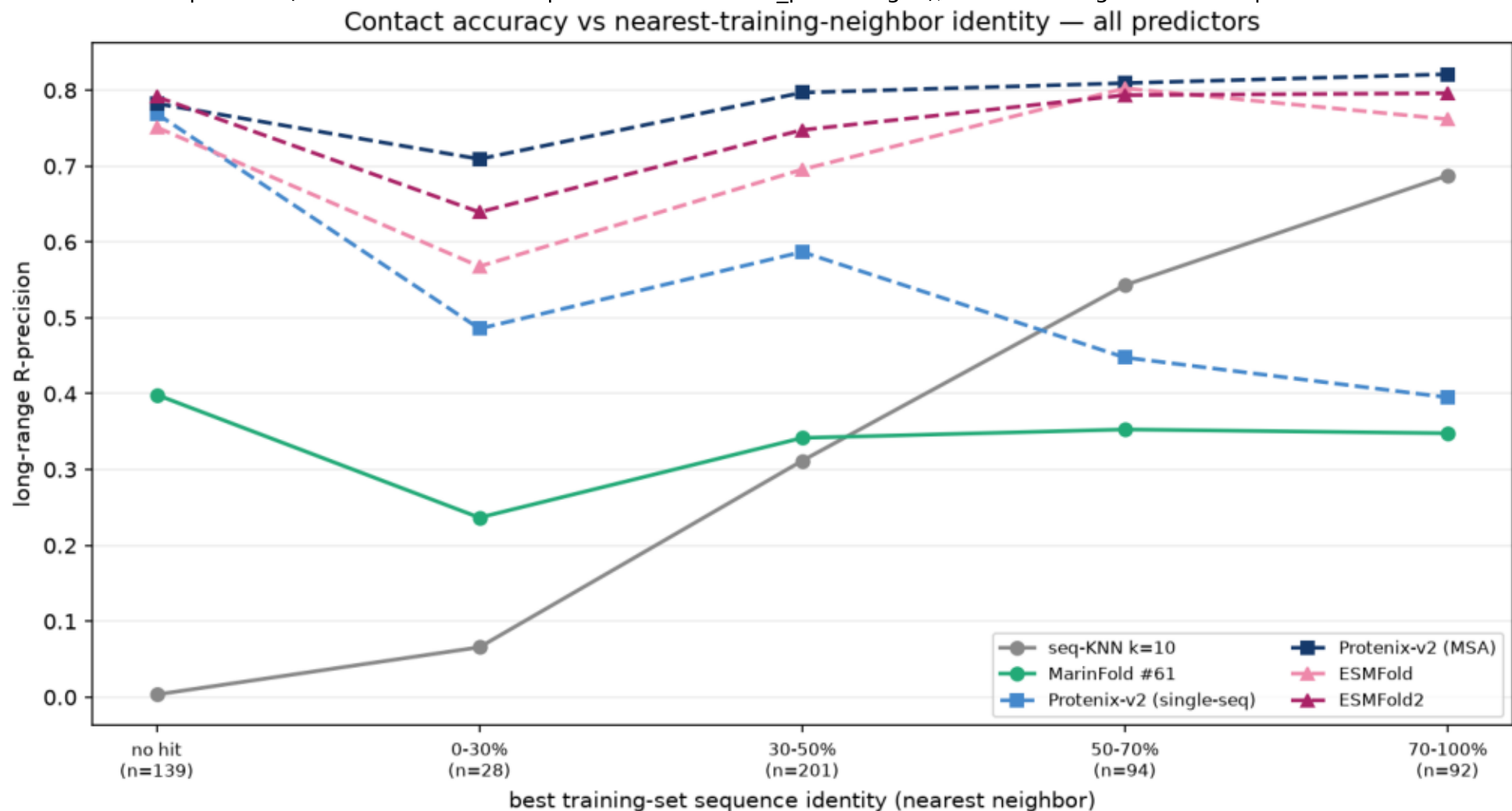
Long-range R-precision binned by best training-set sequence identity ('no hit' = no mmseqs homolog). The core memorization read: seq-KNN k=10 climbs 0.00 to 0.69 while Marifold #61 stays flat (~0.34-0.40), including on the 139 no-homolog proteins.



```
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```

rprecision_vs_identity_all.png

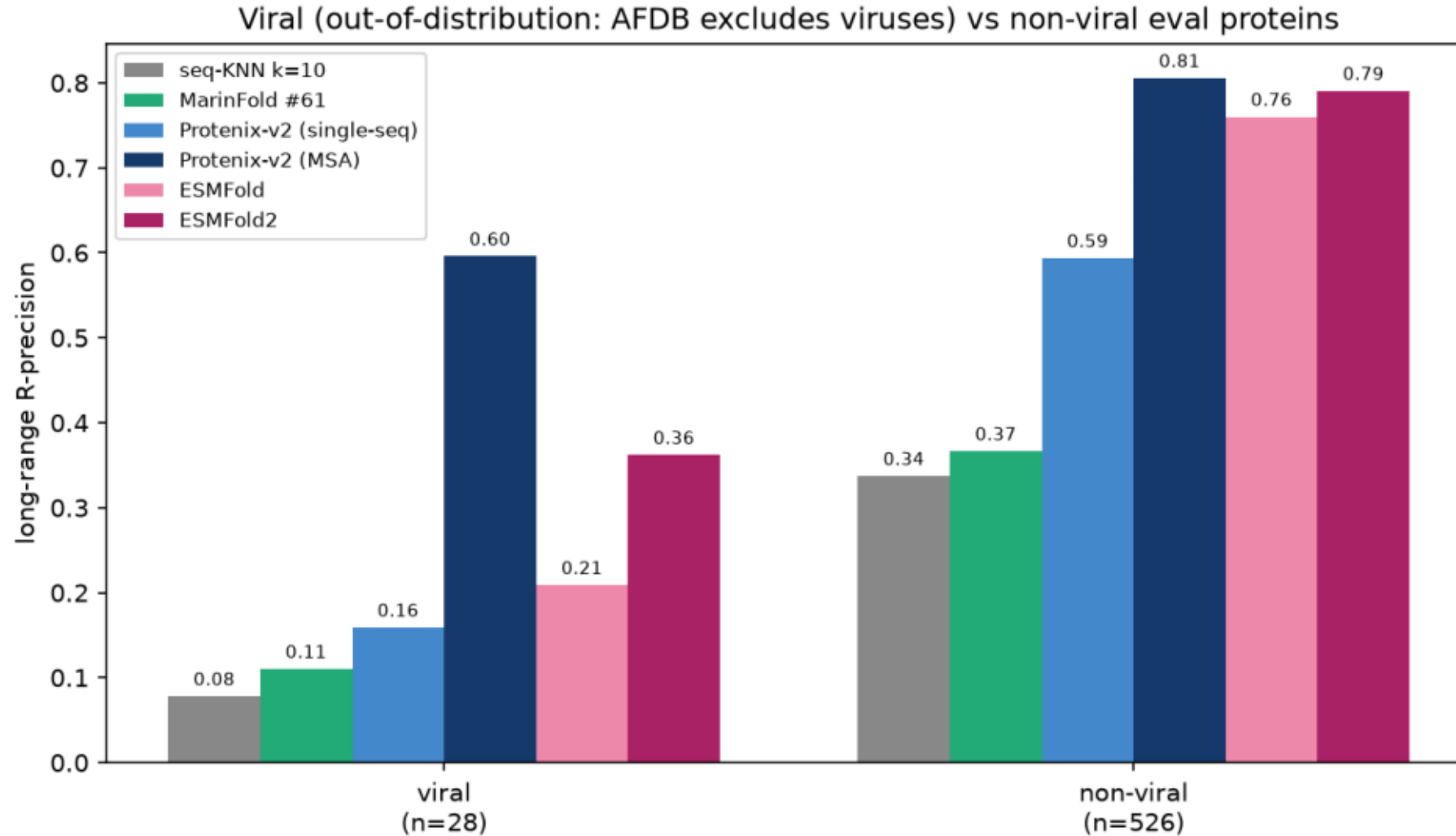
Same identity bins, one line per predictor: seq-KNN is the only one whose accuracy tracks training-sequence identity. Caveat — the axis is confounded with composition (123 of the 139 no-hit proteins are denovo_pdb designs); the robust signal is the slope.



```
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```


viral_vs_nonviral.png

Long-range R-precision on the 28 viral vs 526 non-viral eval proteins. AFDB (the training source) excludes viruses, so viral is out-of-distribution: MarFold drops 3.3x (0.37 to 0.11) and ESMFold2 0.79 to 0.36, while MSA-based Protenix-v2 retains ~74%.



```
$ plot_knn.py --base-csv ../exp82_evals_contacts_v1_contact_prediction/_scratch/contact_precision_with_rollout.csv
```